

Health, Habits, and Cancer: Examining Risk of Cancer Using Support Vector Machines

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Introduction

- Problem: Can a medical history of cancer be predicted based on demographic and health data?
- Approach: Support Vector Machines (SVMs) – linear, radial, polynomial kernels
- Dataset: National Health Interview Survey
 - 35,000 observations with demographic information, health metrics, health habits, history of disease¹

Theoretical Background

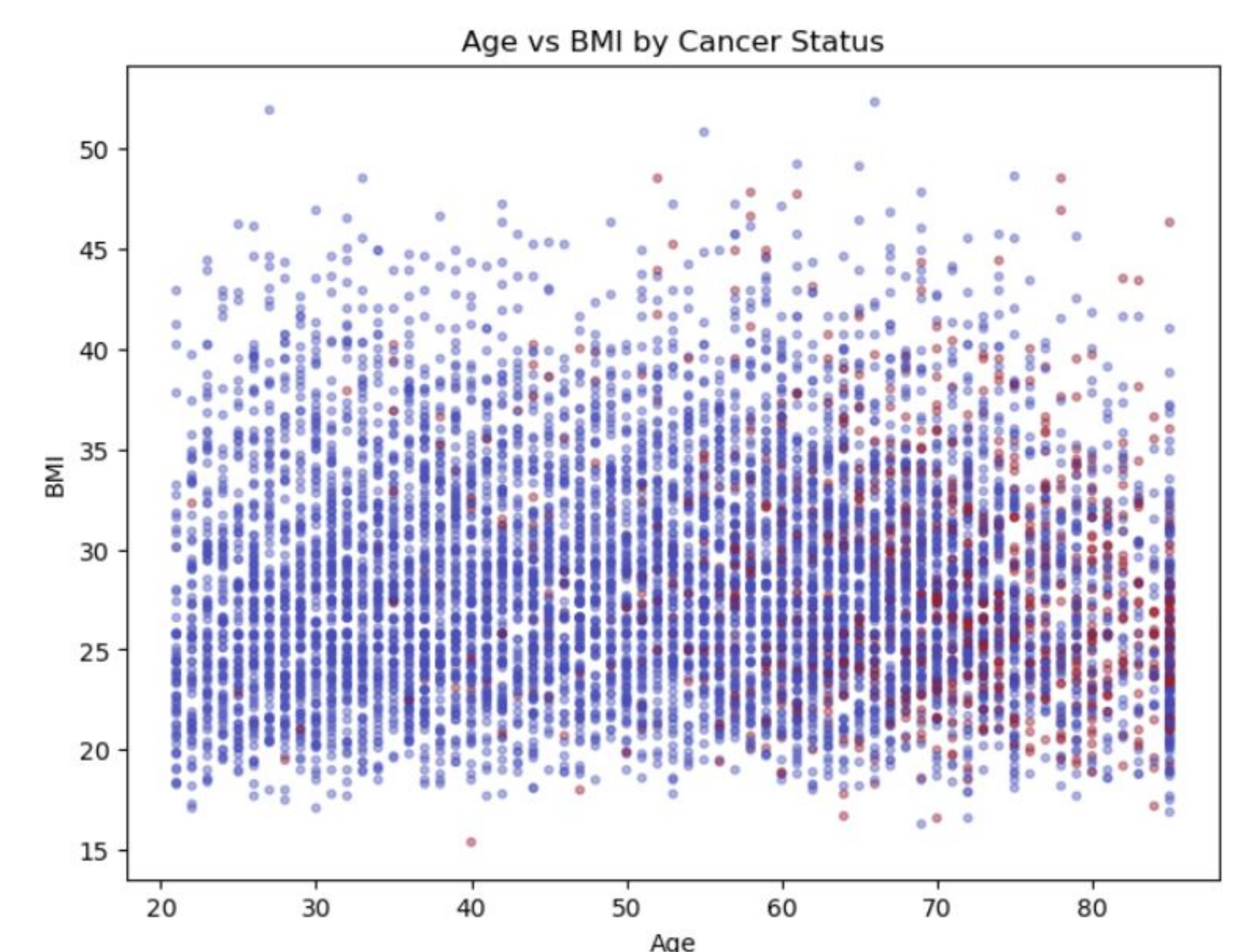
- SVMs find the optimal boundary between two classes by maximizing the margin between this boundary and the nearest data points (support vectors)
- Size of margin controlled by cost parameter
- Key hyperparameters: cost (all models), gamma (radial), degree (polynomial)
- Evaluation metrics: accuracy/error rate, ROC–AUC

Methodology

- Dataset reduced to 7,571 observations (21+, American South)
- Target variable: medical history of cancer (yes or no)
- Predictor variables: age, income level relative to poverty level, BMI, health insurance status, alcohol consumption frequency, duration of moderate physical activity, usual hours of sleep
- 80/20 train–test split; GridSearchCV for each hyperparameter

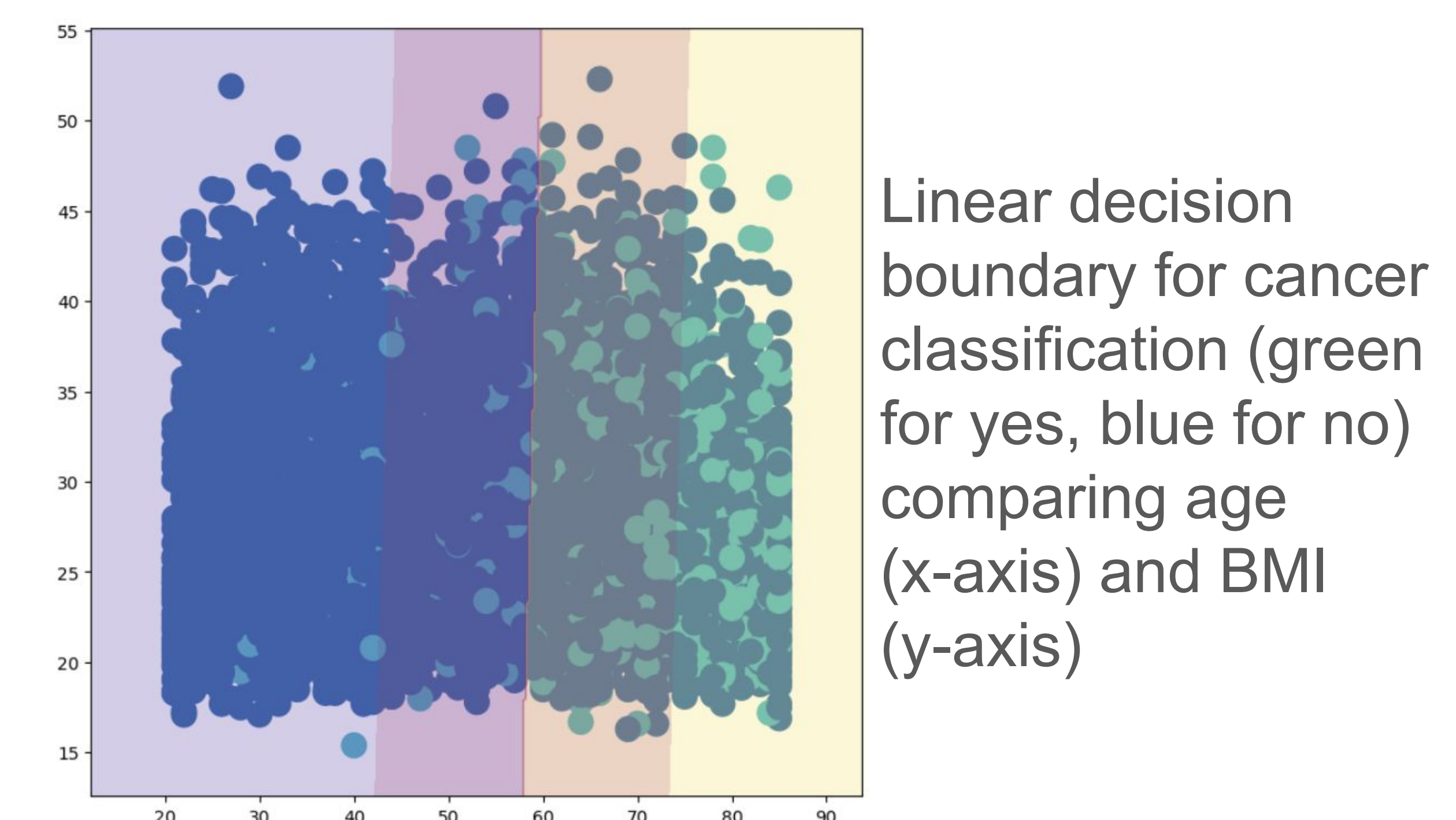
Results

Kernel Type	Test Error Rate (post-tuning)	AUC Score	Optimal Hyperparameters
Linear	33.73%	0.79	cost = 5
Radial	37.56%	0.79	cost = 1; gamma = 0.0001
Polynomial	31.82%	0.79	cost = 0.1; degree = 2



Conclusions

- Age is strongest predictor of history of cancer
- Similar model performance suggests limitation of predictors
- Screening for cancer at specific ages is recommended



[1] Lynn A. Blewett et al. IPUMS Health Surveys: NHIS, Version 7.4. Minneapolis, MN: IPUMS, 2024. <https://doi.org/10.18128/D070.V7.4>